

## CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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## Case 5-2026: An 18-Year-Old Woman with Headache and Hypertension

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## PRESENTATION OF CASE

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CME



*Dr. Darius T. White* (Medicine–Pediatrics): An 18-year-old woman presented to the emergency department of this hospital because of worsening headache and hypertension.

The patient had been in her usual state of health until 6 years before this presentation, when she began to note intermittent headaches, which her primary care physician diagnosed as both tension and migraine headaches. The headaches resolved after the administration of nonsteroidal antiinflammatory drugs (NSAIDs).

One year before the current presentation, the patient had worsening daily headaches that usually occurred in the morning and, at times, woke her up from sleep. The headaches were described as bitemporal pain and throbbing with associated nausea, photophobia, and scintillating scotoma. The patient was referred to the pediatric neurology clinic of this hospital for evaluation.

Evaluation of the patient in the neurology clinic was notable for a blood pressure of 164/100 mm Hg, which was attributed to a headache at the time of the blood pressure reading. The gait and mental status were normal, and no motor or sensory deficits were present. The funduscopic examination showed sharp optic disks in both eyes; however, given the patient's medical history, increased intracranial pressure was clinically suspected. Magnetic resonance imaging (MRI) of the head was ordered, and a formal ophthalmologic examination was recommended, but neither was pursued at that time. Treatment with propranolol was initiated and helped to alleviate her symptoms, but it was discontinued after 2 months owing to insomnia and depression. Amitriptyline was prescribed, which also provided partial relief.

Three weeks before the current presentation, the patient was evaluated in the pediatric neurology clinic of this hospital for worsening headaches that were no longer responsive to therapy. A new symptom, morning nausea with retching, was also noted. A urine pregnancy test was negative, and the symptoms were attributed by the patient to gastroesophageal reflux. The blood pressure was 130/72 mm Hg, which was consistent with recent blood pressure readings, and the neurologic

examination was again normal, although the fundi could not be well visualized at the time of examination. Owing to the escalating frequency and severity of her headaches and the ongoing clinical concern for increased intracranial pressure, topiramate was prescribed, MRI of the head was reordered, and the patient was again urged to undergo an ophthalmologic examination.

MRI of the head, performed after the administration of intravenous gadolinium-based contrast material, showed no evidence of infarction, thrombosis, hemorrhage, masses, or elevation of the optic nerve head. However, a funduscopy examination, performed after pupillary dilation at a local eye clinic, reportedly revealed papilledema. The patient subsequently underwent a lumbar puncture, which was notable for an opening pressure of 36 cm of water (normal range, 10 to 20) and a closing pressure of 10 cm of water after removal of 10 ml of cerebrospinal fluid (CSF). Analysis of the CSF showed normal cell counts and a normal total protein level. A diagnosis of idiopathic intracranial hypertension was made, and treatment was switched from topiramate to acetazolamide.

A few days later, the patient presented to the pediatric emergency department of this hospital with an intensifying and debilitating headache. The headache was bitemporal with crushing pain and associated nausea and photophobia, and it was refractory to the administration of sumatriptan, ondansetron, and naproxen. The blood pressure was 207/143 mm Hg. An anteroposterior chest radiograph showed prominence of the perihilar vasculature with perihilar opacities in both lungs and peribronchial cuffing consistent with pulmonary edema. Treatment with lactated Ringer's solution, intravenous magnesium sulfate, and acetaminophen was initiated, and the headache abated. Owing to severe hypertension, a continuous infusion of nicardipine was initiated, and invasive hemodynamic monitoring was performed with the use of an intraarterial catheter placed in the left radial artery. The patient was admitted to the pediatric intensive care unit.

The oral temperature was 36.1°C, the blood pressure 170/74 mm Hg, the heart rate 102 beats per minute, the respiratory rate 22 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The weight was 110 kg, and the body-mass index (the weight in kilograms divided by the square of the height in

meters) was 36.4. She appeared fatigued but was in no acute distress. No jugular venous distention was present, and auscultation of the heart and lungs revealed no abnormalities. There were no abdominal bruits or palpable lymphadenopathy. The legs were warm, with palpable distal pulses, and no edema was present. A neurologic examination showed no motor or sensory deficits.

The patient's medical history was notable for polycystic ovarian syndrome, migraine, and tension headache. Medications included amitriptyline, norgestimate–ethinyl estradiol, and acetazolamide. In addition, naproxen, ondansetron, and sumatriptan were administered as needed for symptoms of headache. Although the dose of amitriptyline had been increased 3 weeks before this evaluation, treatment with this drug was discontinued because of xerostomia. She had no history of allergy to medications. Her family history was notable for inflammatory bowel disease in her two brothers, who were twins, but there was no history of hypertension or heart disease. The patient was a high school student who had been accepted to college, and she resided in an urban area of New England with her parents and siblings. She did not use over-the-counter decongestants, supplements, alcohol, tobacco, or recreational drugs.

The plasma potassium level was 2.0 mmol per liter (normal range, 3.4 to 5.0), and the initial plasma creatinine level was 1.19 mg per deciliter (105  $\mu$ mol per liter; reference range, 0.50 to 1.50 mg per deciliter [53 to 133  $\mu$ mol per liter]), with improvement to 0.87 mg per deciliter (77  $\mu$ mol per liter) after the administration of intravenous fluids; the baseline value had been reported at 0.60 mg per deciliter. Additional laboratory data are shown in Table 1. Urinalysis showed 2+ proteinuria and a specific gravity of 1.001, with 640 mg of total protein excreted over 24 hours. The thyrotropin level was mildly elevated at 5.99 mIU per liter (reference range, 0.40 to 5.00), with normal free thyroxine levels. The plasma metanephrine and normetanephrine levels were within normal limits. The serum complement levels were normal, and antinuclear antibodies were not detected. Urine toxicology testing was negative for cocaine and amphetamines.

A 12-lead electrocardiogram (Fig. 1) was notable for sinus tachycardia with inferoapical ST-segment depressions and prominent voltages, findings suggestive of left ventricular hypertrophy,

**Table 1. Laboratory Data.\***

| Variable                               | Reference Range, Adults† | On Initial Presentation |
|----------------------------------------|--------------------------|-------------------------|
| Hemoglobin (g/dl)                      | 12.0–16.0                | 15.1                    |
| Hematocrit (%)                         | 36.0–46.0                | 42.5                    |
| White-cell count (per $\mu$ l)         | 4500–13,000              | 10,120                  |
| Platelet count (per $\mu$ l)           | 150,000–450,000          | 350,000                 |
| Sodium (mmol/liter)                    | 135–145                  | 132                     |
| Potassium (mmol/liter)                 | 3.4–5.0                  | 2.0                     |
| Chloride (mmol/liter)                  | 98–108                   | 90                      |
| Carbon dioxide (mmol/liter)            | 23–32                    | 26                      |
| Urea nitrogen (mg/dl)                  | 8–25                     | 16                      |
| Creatinine (mg/dl)                     | 0.50–1.50                | 1.19                    |
| Aspartate aminotransferase (U/liter)   | 9–32                     | 25                      |
| Alanine aminotransferase (U/liter)     | 7–33                     | 23                      |
| Alkaline phosphatase (U/liter)         | 45–87                    | 127                     |
| Total bilirubin (mg/dl)                | <1.2                     | 0.5                     |
| High-sensitivity troponin T (ng/liter) | 0–9                      | 43                      |
| Erythrocyte sedimentation rate (mm/hr) | 0–19                     | 48                      |
| C-reactive protein (mg/liter)          | <8                       | 44                      |
| Thyrotropin (mIU/ml)                   | 0.4–5.0                  | 5.99                    |

\* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

as well as the presence of a U wave. Transthoracic echocardiography showed moderate-to-severe left ventricular hypertrophy with normal systolic function. Ultrasonography of the kidneys and renal vasculature showed normal parenchyma and normal velocities and resistive indexes on both sides.

Owing to persistent hypertension, a continuous infusion of phentolamine was started in addition to treatment with oral labetalol. The patient had refractory hypokalemia despite the use of daily potassium supplementation; ultimately, treatment was started with spironolactone, lisinopril, and losartan, which was associated with improvement in serum potassium levels.

A diagnostic and therapeutic procedure was performed.

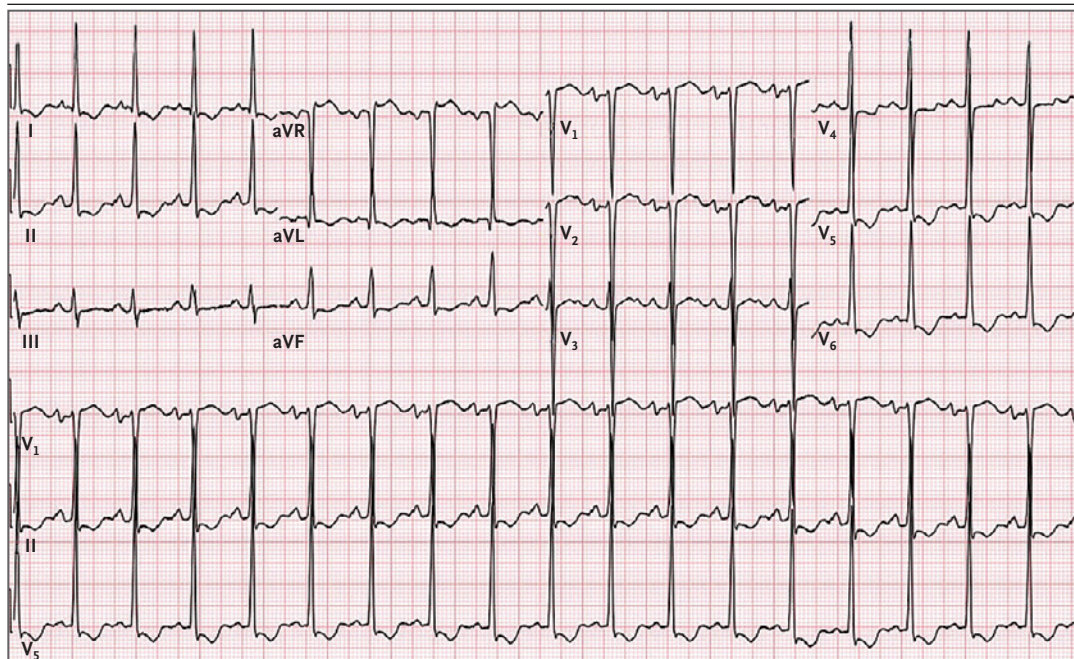
#### DIFFERENTIAL DIAGNOSIS

*Dr. Paul K. Fadakar:* I participated in the care of this patient and am aware of the final diagnosis. This young woman presented with debilitating

headaches, presumably due to refractory hypertension. The differential diagnosis for hypertension in an adolescent includes renal parenchymal disease, from both acute and chronic causes, and extrarenal conditions such as endocrine diseases, medication use, and congenital vascular anomalies.

#### RENAL PARENCHYMAL DISEASES

Common causes of secondary hypertension in children and adolescents are acute kidney injury and chronic kidney disease. Possibilities include acute pyelonephritis, which can result from vesicoureteral reflux; glomerulonephritis, which can occur after streptococcal infection; and reflux nephropathy, which is kidney scarring that can be caused by any chronic renal parenchymal disease. However, this patient had no history of kidney disease. The initial laboratory evaluation was notable for a blood creatinine level of 1.19 mg per deciliter, which improved quickly to 0.87 mg per deciliter, effectively ruling out acute kidney



**Figure 1. Electrocardiogram.**

A 12-lead electrocardiogram shows sinus tachycardia with inferolateral ST-segment depressions and T-wave inversions, findings that indicate subendocardial ischemia or ventricular hypertrophy. Also visible is prominent R-wave amplitude in lead V<sub>6</sub>, S-wave amplitude in lead V<sub>1</sub>, and an R wave in aVL (>11 mm in amplitude), all of which are findings suggestive of left ventricular hypertrophy. The U waves noted in V<sub>2</sub>, V<sub>3</sub>, and V<sub>4</sub> are consistent with severe hypokalemia.

failure as the cause of her elevated blood pressure. Furthermore, no hematuria was present, which also ruled out acute glomerulonephritis as a diagnosis.

#### EXTRARENAL DISEASES ASSOCIATED WITH HYPERTENSION

Extrarenal causes of secondary hypertension include primary hyperaldosteronism, which is one of the most common causes of treatment-resistant hypertension in children and adolescents. However, hypokalemia is present in only approximately one third of patients with primary hyperaldosteronism. In children and adolescents with primary hyperaldosteronism, bilateral idiopathic adrenal hyperplasia is more common than a single aldosterone-producing adenoma (Conn's syndrome) and is associated with elevated aldosterone levels and suppressed renin levels.

In young women with early-onset, accelerated, or resistant hypertension, it is important to consider fibromuscular dysplasia, the incidence of which may be as high as 7.5% among women with hypertension who are younger than 50 years

of age. Although hypokalemia is not a primary feature of this condition, secondary hyperaldosteronism from renal hypoperfusion can occur. Although computed tomographic (CT) angiography is the standard imaging test for the diagnosis of fibromuscular dysplasia, the patient had normal velocities and resistive indexes on a renal vascular ultrasound, which made renal-artery stenosis a less likely diagnosis.

Hypertension in persons younger than 40 years of age is an indication to perform screening for Cushing's syndrome, which can manifest with physical signs (e.g., purple striae >1 cm in width, central obesity, facial plethora, and proximal muscle weakness) — none of which were present in this patient. In patients with Cushing's syndrome, hypertension and hypokalemia result from the mineralocorticoid effects of cortisol. Screening involves a late-night salivary cortisol test, a 24-hour urinary free cortisol test, or an overnight dexamethasone (1 mg) suppression test.

Liddle's syndrome is a rare monogenic disorder that can lead to secondary hypertension.



This autosomal dominant disorder is caused by variants in epithelial sodium-channel genes (*SCNN1A*, *SCNN1B*, or *SCNN1G*), which result in excessive sodium reabsorption. Patients with Liddle's syndrome, who are usually much younger than this patient, present with hypertension, hypokalemia, and unlike those with primary hyperaldosteronism, characteristically low renin and aldosterone levels. If Liddle's syndrome is suspected on the basis of laboratory testing, genetic testing is essential for confirmation.

Finally, patients with pheochromocytoma can present with features similar to those seen in this patient — namely, severe episodic hypertension associated with headaches and hypokalemia (caused by catecholamine-induced kaliuresis). Screening involves measuring the levels of plasma free metanephrines or 24-hour urinary fractionated metanephrines and normetanephrines. However, the results of serum metanephrine and normetanephrine testing were ultimately normal in this patient, which makes pheochromocytoma unlikely.

#### TOXINS AND MEDICATIONS

This patient's urine toxicology screening showed no evidence of stimulant or cocaine use, and a medication history was notable for sparse use of NSAIDs and no use of over-the-counter decongestants, which are known culprits that can lead to hypertension. Although estrogen-based oral contraceptive medications have been associated with the exacerbation of hypertension, the patient's long-standing history of taking this medication, as well as the dramatic presentation, made this an unlikely cause of her hypertension.

#### CONGENITAL VASCULAR ANOMALIES

Coarctation of the aorta should be considered in adolescents with hypertension. This patient had an echocardiogram that showed no evidence of coarctation of the aorta but was notable for left ventricular hypertrophy, a finding that suggests severe and long-standing hypertension.

At this point in the evaluation, most of the possible causes of this patient's refractory hypertension and hypokalemia had been ruled out, except for primary hyperaldosteronism or primary hyperreninemic hypertension. The latter could be the result of a rare primary tumor of the kidney, such as a reninoma. To rule out these possible

diagnoses, we obtained abdominal and pelvic imaging and measured the plasma renin activity and aldosterone level.

#### DIAGNOSTIC TESTING

*Dr. Ali Pourvaziri:* CT angiography of the abdomen and pelvis showed no evidence of renal-artery stenosis. The patient had bilateral single renal arteries originating from the abdominal aorta without radiologically significant narrowing (Fig. 2A). A hypodense lesion was noted in the interpolar region of the left kidney, with a more hypodense focus along its superior margin. The lesion showed low-level enhancement on imaging obtained after the administration of intravenous contrast material (Fig. 2B and 2C). MRI, obtained after the administration of intravenous gadolinium-based contrast material, was performed for further characterization and revealed a lesion with a predominantly hypointense signal on T1- and T2-weighted imaging and a cystic component measuring 8 mm in diameter (Fig. 2D and 2E). Many of the imaging findings in this patient are nonspecific and have features that overlap with many small renal masses, such as papillary renal-cell carcinoma, reninoma, and angiomyolipoma without fat, among others.

*Dr. Thomas Denize:* On hospital day 4, the plasma level of renin was severely elevated at 150 ng per milliliter per hour (reference range, 1.2 to 2.4). The aldosterone level was 4.1 ng per deciliter (reference range, <21).

#### ADDITIONAL DIFFERENTIAL DIAGNOSIS

*Dr. Fadakar:* The fact that this patient had an aldosterone level at the lower limit of the normal range was surprising. Given the high renin level, we expected the aldosterone level to be elevated as well, but it was not. At the time this testing was performed, the patient was not receiving an angiotensin-converting-enzyme (ACE) inhibitor or an angiotensin-receptor blocker; thus, a probable explanation for this result was an inaccurate result owing to interference with the assay or laboratory error. Furthermore, the presence of hyperreninemia also ruled out monogenic causes of hypertension, which are universally associated with very low plasma renin activity.

Given the extremely high plasma renin activity, refractory hypokalemia, and the presence of a single renal mass, we thought the most likely diagnosis was a reninoma. We consulted the urology service to evaluate this patient for possible surgical resection of the renal mass. Resection was scheduled to occur 6 weeks later to provide time to control this patient's refractory hypertension with medication.

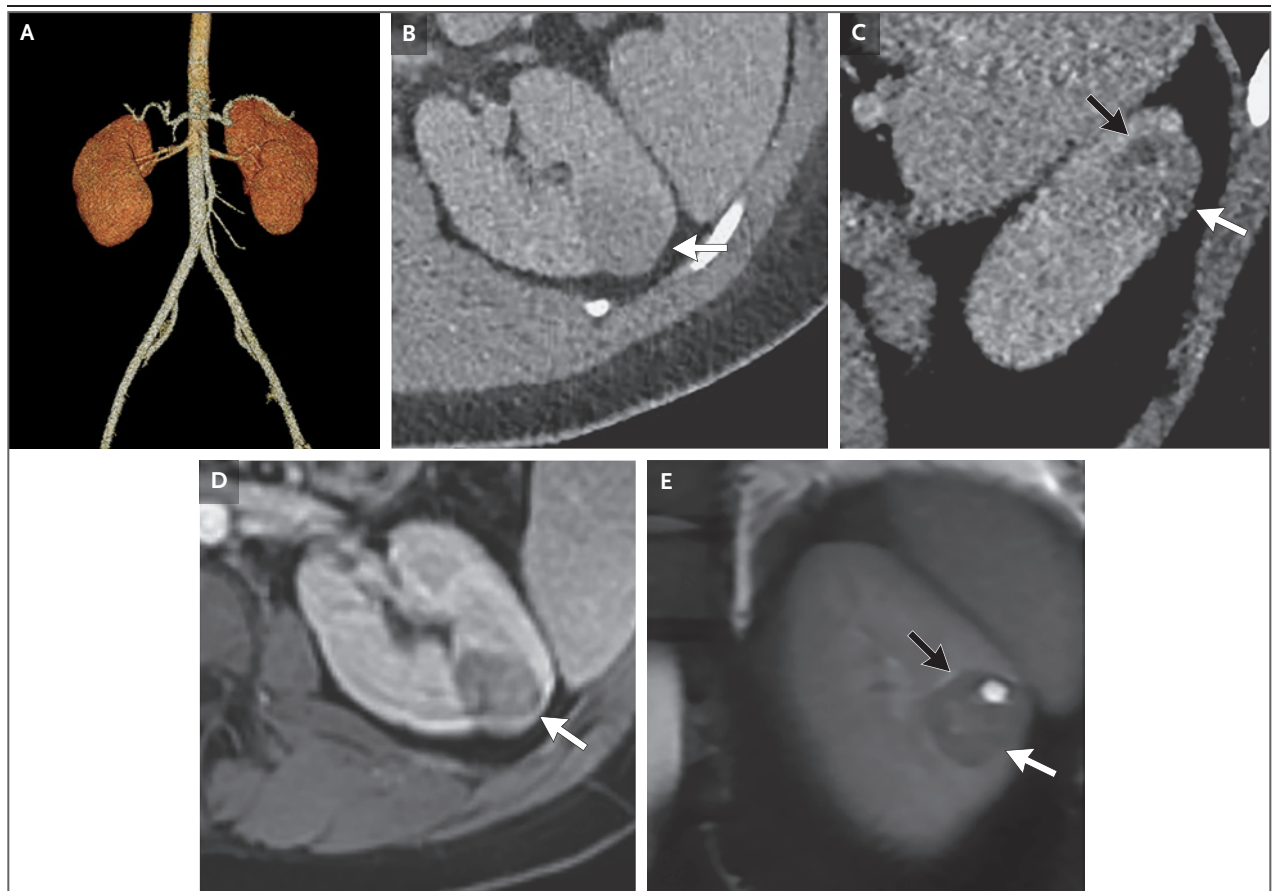
#### DR. PAUL K. FADAKAR'S DIAGNOSIS

Hyperreninemic hypertension due to reninoma.

#### INITIAL MANAGEMENT AND HOSPITAL COURSE

*Dr. Fadakar:* In the emergency department, the patient's blood pressure was greater than 200/100 mm Hg. After the initial doses of labetalol were administered intravenously, a continuous nicardipine infusion was started, and the patient was admitted to the pediatric intensive care unit. The initial goal of treatment was to lower the systolic blood pressure by 25% over the course of 24 hours.

On hospital day 2, a continuous phentolamine



**Figure 2. Imaging Studies of the Abdomen and Pelvis.**

A three-dimensional volume-rendered image of the abdominal aorta and iliac arteries (Panel A) shows no evidence of stenosis or renal-artery abnormalities. An axial CT image, obtained in the portovenous phase after the administration of intravenous contrast material (Panel B), reveals a hypoenhancing, partially exophytic mass in the left kidney (arrow). A sagittal image obtained from contrast-enhanced CT in the portovenous phase (Panel C) shows the same lesion (white arrow), with a more hypodense focus along its superior margin (black arrow). An axial T1-weighted MRI, obtained after the administration of intravenous contrast material (Panel D), shows a hypointense, hypoenhancing lesion in the left kidney (arrow). A coronal T2-weighted MRI (Panel E) shows the lesion with predominantly hypointense signal (white arrow) and a small superior cystic component (black arrow).

infusion was added to the patient's treatment regimen. This alpha-adrenergic inhibitor was chosen while the results of plasma metanephrine testing were pending in order to limit beta-blocker use. In the context of a possible pheochromocytoma, the use of beta-blockers for antihypertensive therapy may lead to unopposed alpha-adrenergic stimulation, which can exacerbate hypertension. Unfortunately, the addition of this agent did not achieve the new treatment goal that day of lowering the systolic blood pressure closer to 160 mm Hg.

On hospital day 3, spironolactone was added to the treatment regimen for further blood pressure control, and for its potassium-sparing effect, given the patient's persistent hypokalemia despite frequent potassium supplementation. Even with the addition of spironolactone, the systolic blood pressure measurements remained elevated (range, 160 to 170 mm Hg).

Owing to the markedly elevated plasma renin activity, captopril was added to the existing antihypertensive regimen. Captopril was chosen for its relatively short half-life, which would allow for easier dose adjustments. Remarkably, the blood pressures did not improve despite progressive increases in the dose of captopril until a maximal dose was reached. Eventually, treatment was switched from captopril to lisinopril. Losartan, an angiotensin-receptor blocker, was also started to enable tapering and then stopping of the continuous nicardipine infusion. Furthermore, the combination of the angiotensin-receptor blocker with an ACE inhibitor was considered useful to address any additional angiotensin II production that would not be prevented by upstream ACE inhibition.

Despite these interventions, the patient's systolic blood pressure could not be consistently maintained below 150 mm Hg; thus, oral labetalol was added to the treatment regimen, which finally allowed us to discontinue the nicardipine infusion. Although treatment with parenteral antihypertensives was no longer needed, the patient remained hypertensive despite receiving maximal doses of lisinopril, losartan, labetalol, and spironolactone.

Before the patient was discharged, treatment with a direct renin inhibitor, aliskiren, was initiated for additional blood pressure control, given the concern for hyperreninemic hypertension. All other medications were continued at maxi-

mal doses, and she was discharged from the hospital with instructions to monitor blood pressure at home.

#### SURGICAL MANAGEMENT

*Dr. Adam S. Feldman:* Six weeks after the initial hospitalization for refractory hypertension, the patient was admitted for resection of the renal mass. A predominantly solid renal mass measuring 2.9 cm in diameter identified on imaging studies would have a limited number of possible diagnoses. Among adults, the probability of a small renal mass being a renal-cell carcinoma is approximately 80%.<sup>1</sup>

Although renal-cell carcinoma is uncommon in the pediatric population and in young adults, with an age-standardized rate of 0.6 cases per 100,000 person-years, it still should remain high on the differential diagnosis of a small renal mass.<sup>2</sup> In this patient, who presented with hypertensive crisis and elevated plasma renin activity, a juxtaglomerular cell tumor, also known as reninoma, was highest on the differential diagnosis.

In patients with a small renal mass, an imaging-guided percutaneous core-needle biopsy has a diagnostic accuracy of approximately 90% and is associated with a low incidence of adverse events.<sup>3</sup> The American Urological Association guidelines on the management of small renal masses recommend consideration of a renal mass biopsy with a "utility-based approach," such that it is used when the results will influence management.<sup>4</sup> In this 18-year-old woman who is symptomatic from this small, hormonally active renal mass, treatment is indicated; therefore, pursuing a renal mass biopsy would not have influenced the management of this patient's renal mass. Furthermore, even in an 18-year-old woman with a small renal mass who is asymptomatic, indefinite lifelong surveillance imaging would be required even if the biopsy results suggested a benign lesion; therefore, avoiding this diagnostic procedure and proceeding directly to surgical resection was felt to be reasonable in this case.

Resection of the mass was performed with the use of minimally invasive robot-assisted laparoscopic partial nephrectomy. Partial nephrectomy is an approach in which the tumor is removed with maximal preservation of the normal renal parenchyma. This method is used in most cases of

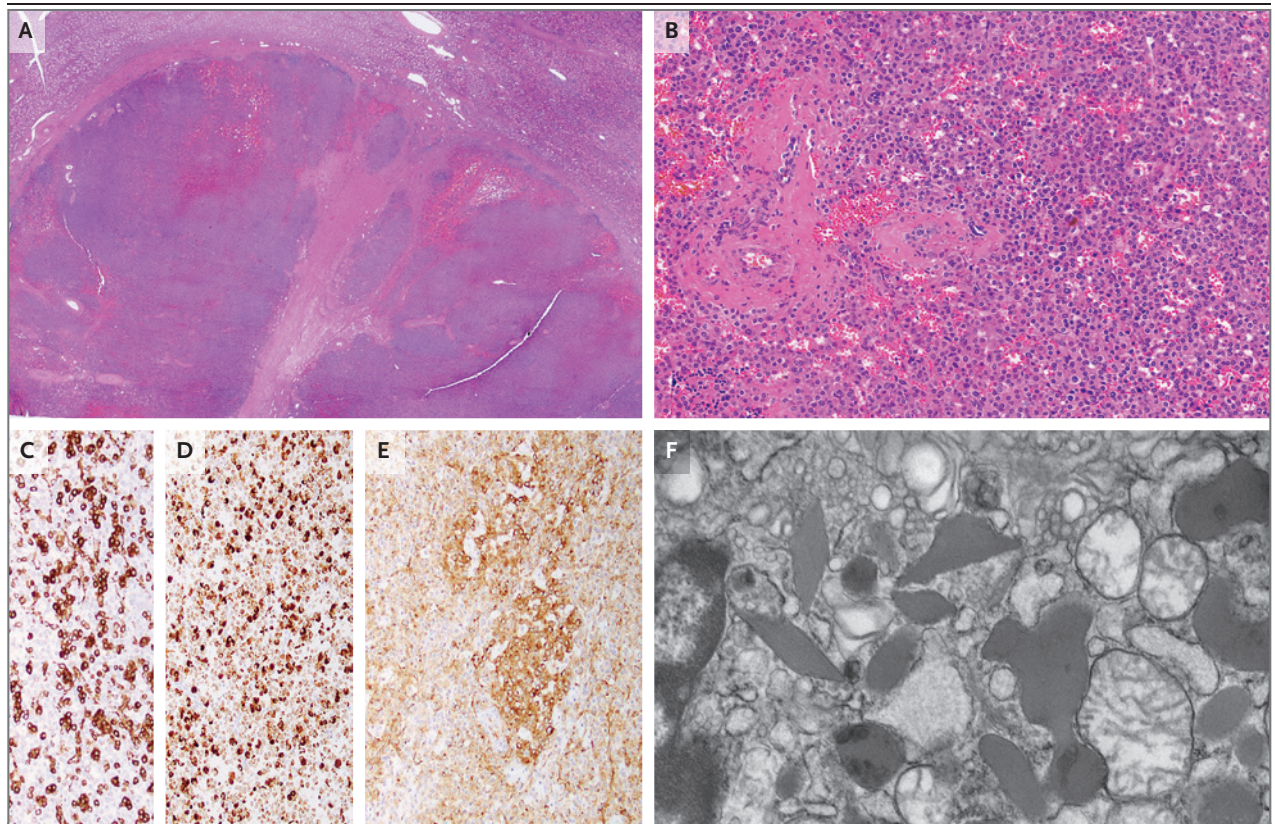


clinical T1a small ( $\leq 4$  cm) renal masses and selected larger renal masses. The rationale for partial nephrectomy and a nephron-sparing approach is preservation of renal functional reserve, which can decrease the risk of new-onset chronic renal insufficiency and, in young patients, has been suggested to improve overall survival as compared with radical nephrectomy.<sup>5,6</sup> In this patient's case, the robot-assisted laparoscopic partial nephrectomy was performed with a low renal artery clamp time of less than 20 minutes, no intraoperative complications, and no hypotensive or hypertensive events. The patient was discharged home on postoperative day 1 with no complications.

In the days after discharge, treatment with aliskiren was discontinued because of hypotensive episodes at home. Subsequently, the patient remained normotensive without therapy.

## PATHOLOGICAL DISCUSSION

*Dr. Denize:* Hematoxylin and eosin staining of the excised specimen revealed a nodular, well-limited, and encapsulated lesion with fibrous scarring (Fig. 3A) and homogeneous proliferation of small cells with poorly visible cell borders and hyalinized blood vessels (Fig. 3B). These findings are histologically consistent with a reninoma, which is characterized by sheets of small polygonal-to-spindle cells that have indistinct cell borders, round nuclei, and eosinophilic-to-clear cytoplasm.<sup>7</sup> The vasculature can be prominent, with hyalinized blood vessels. Entrapped normal renal tubules can be present at the periphery of the lesion. The majority of these tumors are benign, with a few case reports involving patients with locally recurrent or metastatic disease.<sup>8-10</sup> Some proposed



**Figure 3. Excisional Biopsy Specimen of the Renal Mass.**

Hematoxylin and eosin staining reveals a nodular, well-limited, and encapsulated lesion with fibrous scarring (Panel A) and, at a higher magnification, homogeneous proliferation of small cells with poorly visible cell borders and hyalinized blood vessels (Panel B). Immunostaining for smooth-muscle actin (Panel C), GATA3 (Panel D), and CD34 (Panel E) is positive, a finding that supports a diagnosis of reninoma. An image from electron microscopy (Panel F) reveals rhomboid renin crystals.



adverse histologic features that are potentially predictive of a poorer prognosis include vascular invasion, necrosis, and infiltrative growth.

The most specific immunohistochemical study for reninoma, although seldom available, is for renin, which is positive in patients with reninoma, with positive control in the normal juxtaglomerular apparatus. Other helpful immunohistochemical studies for the diagnosis of reninoma include those for smooth-muscle actin (Fig. 3C), GATA3 (Fig. 3D), and CD34 (Fig. 3E), all of which were positive in this patient.<sup>11</sup> In addition, on electron microscopy (Fig. 3F), renin crystals (also known as renin rhomboid protogranules) were detected. No recurrent mutation has been identified for the diagnosis of reninoma. Likewise, no recurrent gene fusions are known to distinguish reninomas from glomus tumors, which may be considered in the differential diagnosis. However, recurrent losses of chromosomes 9 and 11 have been reported in some studies.<sup>12,13</sup>

#### PATHOLOGICAL DIAGNOSIS

Reninoma.

#### FOLLOW-UP

*Dr. Danielle B. Pier:* Between the hospitalization for hypertensive crisis and the planned admission for removal of the renal mass 6 weeks later, the patient had frequent follow-up visits in the neuro-ophthalmology clinic. During these visits, she was normotensive but continued to report headache with pulsatile tinnitus. An ophthalmologic examination showed papilledema in both eyes (Frisén grade 2 to 3 in the right eye and grade 2 in the left), as well as macular blot hemorrhages in both eyes and a cotton-wool spot in the left eye. Perimetry showed mild visual field loss in the bilateral nasal fields. Given these findings, her clinical team suspected ongoing elevation of intracranial pressure. It was unclear whether this process was related to the renin-secreting tumor or was a separate process of idiopathic intracranial hypertension. A lumbar puncture that was performed 3 weeks after hospital discharge revealed an opening pressure of 32 cm of water, a finding indicative of ongoing elevated intracranial pressure.

Links between endocrine abnormalities and increased intracranial pressure have been suspected since the 1950s.<sup>14,15</sup> A case series describing 12 patients with hyperaldosteronism and idiopathic intracranial hypertension reported presenting symptoms of chronic headaches, visual disturbances, hypokalemia, hypomagnesemia, and metabolic alkalosis in all 12 patients, regardless of the cause.<sup>16</sup> It has been proposed that when aldosterone stimulates the mineralocorticoid receptor in the choroid plexus epithelium, downstream effects include increased synthesis of the sodium-potassium-ATPase pump, which is a primary driver of CSF production.<sup>17</sup> It was concluded that this patient's ongoing hyperreninemia caused a hyperaldosteronism-like environment, despite the finding of normal serum aldosterone levels, and led to CSF overproduction. It was unclear how long it would take for this process to resolve after the removal of the renin-secreting tumor.

Treatment with acetazolamide had been discontinued during the patient's hospitalization for hypertensive crisis. Given the presence of ongoing papilledema and the risk of visual impairment, treatment with acetazolamide was restarted. Perimetry testing 3 weeks after surgery showed improvement, with resolution of the previously noted nasal-field deficits. Papilledema remained present, and treatment with acetazolamide was continued.

Two months after surgery, the patient reported an average of three mild headaches per month, all of which resolved without medication. She remained normotensive. Her only regular medication was acetazolamide. Six months after surgery, the patient had completely normal visual fields on perimetry testing. The papilledema had abated but was not completely resolved, and treatment with acetazolamide was continued. Other than an occasional mild headache, the symptoms resolved, and she returned to normal activities. The patient graduated from high school and enrolled in college later that year.

#### FINAL DIAGNOSIS

Reninoma.

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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